

## C-Peptide

### Test Details

#### METHODOLOGY

Electrochemiluminescence immunoassay (ECLIA)

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#### RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

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### Use

The principal use of C-peptide is in the evaluation of hypoglycemia. Patients with insulin-secreting neoplasms have high levels of both C-peptide and endogenous insulin; in contrast, patients with factitious hypoglycemia will have low C-peptide levels in the presence of elevated (exogenous) serum insulin. C-peptide is also useful in evaluating residual beta-cell function in insulin-dependent diabetics, many of whom have antibodies that interfere with insulin assays. Glucagon-stimulated C-peptide concentration has been shown to be a good discriminator between insulin-requiring and non-insulin-requiring diabetic patients. The diagnosis of islet cell tumor is supported by elevation of C-peptide when plasma glucose is low.

## Special Instructions

Values obtained with different assay methods should not be used interchangeably in serial testing. It is recommended that only one assay method be used consistently to monitor each patient's course of therapy. This procedure does not provide serial monitoring; it is intended for one-time use only. If serial monitoring is required, please use the serial monitoring number **480108** (/tests/480108/c-peptide-serial-monitor) to order.

This test may exhibit interference when sample is collected from a person who is consuming a supplement with a high dose of biotin (also termed as vitamin B7 or B8, vitamin H, or coenzyme R). It is recommended to ask all patients who may be indicated for this test about biotin supplementation. Patients should be cautioned to stop biotin consumption at least 72 hours prior to the collection of a sample.

## Limitations

C-peptide levels are increased with renal failure. (C-peptide is normally excreted by the kidneys.) Instances of insulinoma have been described in which proinsulin was increased but insulin and C-peptide were not.

As with all tests containing monoclonal mouse antibodies, erroneous findings may be obtained from samples taken from patients who have been treated with monoclonal mouse antibodies or who have received them for diagnostic purposes.<sup>1</sup> In rare cases, interference due to extremely high titers of antibodies to streptavidin and ruthenium can occur.<sup>1</sup> The test contains additives, which minimize these effects.

## Custom Additional Information

This immunoassay is intended for the in vitro quantitative determination of C-peptide in human serum, plasma, and urine. The assay is intended for use as an aid in the diagnosis and treatment of patients with abnormal insulin secretion.

C-peptide is a single chain 31-amino acid (AA 33-63) connecting (C) polypeptide with a molecular weight of approximately 3021 daltons.<sup>2,3</sup> In the process of biosynthesis of insulin, the C-peptide is formed as a byproduct together with insulin by the proteolytic cleavage of the precursor molecule proinsulin, stored in secretory granules in the Golgi complex of the pancreatic  $\beta$ -cells. Proinsulin, in turn, was cleaved from preproinsulin.<sup>3,4</sup>

C-peptide fulfills an important function in the assembly of the two-chain insulin ( $\alpha$ - and  $\beta$ -chain) structure and the formation of the two disulfide bonds within the proinsulin molecule. Insulin and C-peptide are secreted in equimolar amounts and released into circulation via the portal vein.<sup>5</sup> As half of the insulin, but almost none of the C-peptide, is extracted in the liver, C-peptide has a longer half-life (about 35 minutes) than insulin; 5 to 10 times higher concentration of C-peptide persist in the peripheral circulation, and these levels fluctuate less than insulin.<sup>3-5</sup>

The liver does not extract C-peptide, which is removed from the circulation by the kidneys and degraded, with a fraction excreted unchanged in the urine. The concentration in urine is about 20- to 50-fold higher than in serum. C-peptide concentrations are, therefore, elevated in renal disease.<sup>2-4</sup>

In the past, C-peptide has been considered biologically inactive; however, recent studies have demonstrated that it is capable of eliciting molecular and physiological effects suggesting that C-peptide is in fact a bioactive peptide. There is evidence that C-peptide replacement, together with insulin administration, may prevent the development or retard the progression of long-term complications in type 1 diabetes.<sup>6-11</sup>

Measurements of C-peptide, insulin, and glucose are used as an aid in the differential diagnosis of hypoglycemia (factitious hypoglycemia and hypoglycemia caused by hyperinsulinism) to ensure an appropriate management and therapy of the patients. To quantify the endogenous insulin secretion, C-peptide is measured basally, after fasting and after stimulation and suppression tests. Due to high prevalence of endogenous anti-insulin antibodies, C-peptide concentrations reflect the endogenous pancreatic insulin secretion more reliably in insulin-treated diabetics than the levels of insulin itself. Measurements of C-peptide may, therefore, be an aid in the assessment of a residual  $\beta$ -cell function in the early stages of type-1 diabetes mellitus and for the differential diagnosis of latent autoimmune diabetes of adults (LADA) and type-2 diabetes.<sup>3,4,12-15</sup>

C-peptide measurements are also used to assess the success of islet transplantation and for monitoring after pancreatectomy.<sup>3,4</sup>

Urine C-peptide is measured when a continuous assessment of  $\beta$ -cell function is desired or frequent blood sampling is not practical (eg, in children).<sup>3</sup> C-peptide excretion in urine has been used to assess pancreatic function in gestational diabetes, and in patients with unstable glycemic control in insulin-dependent diabetes mellitus (IDDM).<sup>16,17</sup>

Although testing for C-peptide is not requested for the routine monitoring of diabetes, it is a valuable tool for the individual therapeutic decisions which are essential for an optimal long-term metabolic control.<sup>18,19</sup>

Elevated C-peptide levels may result from increased  $\beta$ -cell activity observed in hyperinsulinism, from renal insufficiency, and obesity.<sup>2</sup> Correlation was also found between higher C-peptide levels and increasing hyperlipoproteinemia and hypertension.<sup>20</sup> Decreased C-peptide levels are observed in starvation, factitious hypoglycemia, hypoinsulinism (NIDDM, IDDM), Addison disease, and after radical pancreatectomy.

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# Specimen Requirements

## Specimen

Serum

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## Volume

0.8 mL

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## Minimum Volume

0.3 mL (**Note:** This volume does **not** allow for repeat testing.)

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## Container

Red-top tube or gel-barrier tube

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## Collection Instructions

If a red-top tube is used, transfer separated serum to a plastic transport tube. Avoid hemolysis.

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## Stability Requirements

| Temperature        | Period    |
|--------------------|-----------|
| Room temperature   | 1 day     |
| Refrigerated       | 14 days   |
| Frozen             | 14 days   |
| Freeze/thaw cycles | Stable x3 |

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## Reference Range

1.1–4.4 ng/mL (**Note:** Reference interval is for fasting patients.)

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## Storage Instructions

Refrigerate.

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## Patient Preparation

Patient should fast for 14 to 16 hours for basal values.

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## Causes for Rejection

Citrate plasma specimen

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## Footnotes

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